

Tyler Chisholm

Technical Artist · Real-Time Rendering

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Technical artist with a rendering-engineer background. I've built a hardware ray-traced Vulkan renderer (ReSTIR, SVGF, DLSS Ray Reconstruction) and shipped UE5 features on a team (compute-shader flocking, lighting architecture, Chaos destruction, editor tools). I profile before I optimize and build tools so artists aren't blocked.

Education

M.S. Interactive Entertainment, Florida Interactive Entertainment Academy (FIEA), University of Central Florida. Dec 2026
12 games built across 4 semesters in UE5, Unity and VR.

B.B.A. Finance & Entrepreneurship, Stetson University, DeLand, FL May 2024

Skills

Rendering Vulkan ray tracing, ReSTIR, SVGF denoising, ray-traced GI, PBR, volumetrics, compute shaders, raymarching/SDFs, stencil techniques
Engines Unreal Engine 5 (Nanite, Lumen, VSM, Substrate, Niagara, Chaos), Unity URP, Godot (engine fork)
Languages HLSL, GLSL, C++, C#, Rust, Python
Tools Nsight Graphics, Perforce, Git, Blender, Unreal Python, Jira

Rendering & Tech Art

Spatial Engine — ray-traced engine for non-Euclidean spaces Rust · Vulkan RT · HLSL · OpenXR 2025 – Present

Personal engine, ~145k lines across 13 crates and 600+ commits. I own the architecture, direction and verification; implementation is AI-assisted.

- Hardware ray-traced Vulkan renderer where spaces with different geometry (hyperbolic, spherical, toroidal) connect through portals, with lighting and shadows traced across them.
- Led a measured VR graphics experiment after streamed terrain exposed a radiance-cache bottleneck; extending the distance cascade cut the cache-resolve median from 2.8–5.8 ms to about 0.35 ms.
- Lighting stack: ReSTIR direct lighting, one-bounce diffuse GI, SVGF denoising, physically based sky and volumetric atmosphere, DLSS Ray Reconstruction; analytic intersection shaders for curved primitives.
- Found and fixed a shader performance regression from GPU timestamp traces by replacing a 64-step sphere tracer with an analytic quadratic solve.
- Integrated into a Godot 4.6 fork as a custom renderer behind a C ABI, so Godot supplies the editor, scripting and asset pipeline.

Design Lead — Everent (Studio Kismet, FIEA capstone) UE5.7 · C++ · Blueprint · Chaos · Perforce 2025 – Present

Momentum-based hoverboard game across an open world of floating sky islands in UE5.7 (Nanite, Virtual Shadow Maps, Substrate, World Partition), coming to Steam.

- Designed and built velocity-gated destruction system in C++ on Chaos Geometry Collections: two-volume speed detection plus an impact-spawned physics field (strain, directional and radial velocity). It doubles as secret gating and skill expression.
- Built an editor tool (Window > Destructibles) that lists every destructible in the level, exposes its tuning and fires a test projectile with in-place reset, so designers tune without entering Play mode.
- Prototyped a compute-shader GPU flocking plugin for ambient sky creatures; when Niagara integration failed, pivoted to instanced meshes driven by CPU readback.
- Led the lighting and shadow architecture decision (Lumen vs. baked GI, VSM cost, HLOD) and wrote the team's graphics-settings spec on UGameUserSettings with runtime scalability groups.

Selected Projects

Pocket Spaces — stencil portal system (bigger on the inside) Unity URP · C# 2026

- Doorways reveal rooms larger than their exteriors via stencil masking; a player state machine, per-object portal membership and gravity overrides let the player and physics objects cross cleanly.
- Raymarched hyperbolic Mandelbox volume inside a pocket room with Möbius-transform camera movement; documented stencil depth limits (render queue collapse at depth 3+).

Earlier Experience

Quality Assurance Agent — Stormbreaker Studios, Orlando, FL Jan – Jul 2020

Tested 8 iOS client apps; wrote 85 defect tickets with repro steps and fix recommendations.